

Assignment 2 Worksheet

Framing the Strait of Hormuz Crisis on Social Media: A Cross-Platform Analysis of Reddit and YouTube

Cover information	Details
Course	COSC2671 - Social Media and Network Analysis
Assignment	Assignment 2
Group	Group 3 (PG)
Team members	Matteo Omizzolo (s4230379); Emilija Milanovic (s4233176)
Project period	12 May 2026 - 25 May 2026
Submission date	25 May 2026

Project Aim And Research Question

The project investigated how Reddit and YouTube communities framed the Strait of Hormuz crisis, and how sentiment, topics, influence, community structure and diffusion differed across platforms. The approach combined text analysis with directed, weighted author-reply networks so that discourse patterns could be interpreted alongside interaction structure.

Data Sources And Analytical Scope

- Reddit discussion data: 55 submissions and 17,162 processed English comments across eight subreddits.
- YouTube discussion data: 291 videos and 26,136 processed English comments.
- FRED daily energy-price data: external temporal context for the crisis and market discussion.
- Core analysis: preprocessing, sentiment and topic analysis, reply-network construction, centrality/community/influence analysis, diffusion analysis, visualisation and cross-platform synthesis.

1. Team Project Plan

The work was organised over two weeks with shared responsibility for the analysis and final submission. The team divided operational tasks evenly, while using regular check-ins and peer review to ensure that both members understood the data pipeline, key results and final argument. Decisions affecting interpretation, scope or presentation were made collaboratively.

Planned Workflow

Phase	Work undertaken	Management approach
Scope and setup	Confirm question, comparison framework, repository structure and analysis sequence.	Agree priorities early and keep shared task list current.
Data and preprocessing	Consolidate Reddit/YouTube data; clean comments; check duplicates, language and usable interaction fields.	Share data-quality checks and document exclusions.
NLP analysis	Prepare text; run sentiment and topic/frame analysis; compare platform-level patterns.	Review parameter choices and interpret results jointly.
Network analysis	Build author-reply graphs; compute metrics, communities, influence and diffusion measures.	Cross-check graph definitions and summary tables.
Synthesis and outputs	Prepare visualisations; connect textual and structural findings; draft report and presentation material.	Co-edit argument and verify all cited values.
Final submission	Proofread report; organise readable repository, README, sample data, access file and worksheet.	Use a final checklist and shared approval.

Division Of Responsibilities

The workload was split evenly. Rather than isolating one member from a major analytical component, both members contributed across collection, preprocessing, NLP, network analysis, visualisation, writing, presentation preparation and final review. Day-to-day ownership was distributed to keep work efficient, followed by peer checking and shared integration.

Responsibility area	Shared allocation and review procedure
Data collection and consolidation	Both members supported source collection/consolidation and verified that the platform datasets aligned with the research question.
Cleaning and preprocessing	Cleaning, duplicate/language checks and text preparation were divided between platforms, with checks reviewed together.
NLP/topic/sentiment analysis	Both members contributed to model outputs, summary tables and interpretation of platform differences.
Network construction and metrics	Both members contributed to interaction graph construction, community/influence metrics and interpretation.
Reporting and presentation	Writing, figures and presentation material were jointly revised for consistency and clarity.
Final submission package	Both members checked filenames, repository organisation, documentation, sample data and supporting files.

Team Management And Quality Assurance

Communication Plan

- Maintain regular check-ins during both weeks to review progress, raise small blockers and agree immediate priorities.
- Coordinate notebook, figure and report changes through the shared repository and a clear file structure.
- Discuss analytical choices together before finalising cross-platform comparisons or written conclusions.
- Complete a final joint review of the report, code package, worksheet, access documentation and data sample.

Quality Control Plan

Quality check	Procedure
Data validity	Check usable text, duplicate handling, language filtering and reply-edge construction before analysis.
Reproducibility	Keep notebooks in execution order, document dependencies and provide a representative sample.
Results consistency	Match figures, tables and reported metrics against notebook outputs during drafting and final review.
Submission quality	Proofread writing; check file naming, privacy, access instructions and repository readability.

Risk Management Plan

Potential risk	Practical response
Uneven platform structures or data quality limitations	Report platform-specific limitations clearly and avoid overstating comparisons.
Analysis or interpretation disagreement	Review evidence together, resolve minor differences through discussion and retain the interpretation best supported by results.
Version confusion near submission	Use clear filenames, repository organisation and a final agreed package check.
Sensitive or oversized submission data	Exclude credentials and full data; provide a documented representative sample within the size requirement.

Final Submission Preparation

In the final stage, both members shared responsibility for integrating the code, analysis, report and presentation material. The package review covered readable source/notebooks, a concise README, an access demonstration file, the worksheet, a representative data sample, privacy checks and correct PDF naming before submission.

2. Weekly Team Plan

The two-week schedule kept analytical tasks and reporting aligned, with responsibilities distributed evenly and reviewed collaboratively at each stage.

Week and focus	Main goals and planned tasks	Responsible members	Expected outputs	Communication / check-in
Week 1 12-18 May 2026	Confirm scope and research question; organise repository and workflow; consolidate Reddit/YouTube data; begin cleaning, duplicate and language checks; define NLP and network pipeline; assign activities evenly.	Matteo Omizzolo and Emilija Milanovic; tasks divided evenly with mutual review.	Agreed question and workflow; organised analysis structure; cleaned/prepared data in progress; analysis plan.	Regular check-ins to review data preparation, task balance and analytical definitions.
Week 2 19-25 May 2026	Finalise preprocessing; run sentiment/topic and network analyses; generate figures/tables; interpret results; write and revise report; integrate presentation and final submission documentation.	Matteo Omizzolo and Emilija Milanovic; shared analytical interpretation and final QA.	Final results and visualisations; revised report; submission-ready repository, README, access file, worksheet and sample data.	Frequent progress updates and a final joint consistency/submission review.

Progress Tracking And Coordination

Management practice	How it supported efficient teamwork
Even workload distribution	The team shared the major workstreams and balanced weekly effort, reducing dependence on one person for interpretation or final delivery.
Regular communication	Check-ins supported quick resolution of minor issues and made emerging results visible to both members.
Collaborative decision-making	Scope, analytical interpretation, visual presentation and report revisions were reviewed together.
Final integration	Both members reviewed consistency across notebooks, figures, report content, presentation material and the submission repository.

Note on timesheets: hours below are approximate records prepared on the confirmed basis that the workload was divided evenly. Both members show equal totals across the project period.

3. Weekly Timesheets

Team member: Matteo Omizzolo (s4230379)

Week 1: 12 May 2026 - 18 May 2026

Date range	Tasks completed	Approx. hours	Outputs produced	Collaboration / communication
12-13 May	Confirmed cross-platform scope and contributed to repository/workflow organisation.	2.0	Agreed research question and structured workflow.	Joint planning and task allocation check-in.
14-15 May	Consolidated Reddit/YouTube data structure and reviewed cleaning requirements.	2.5	Data-field and preprocessing review notes.	Shared decisions on usable text and interaction fields.
16-17 May	Worked on comment cleaning, duplicate/language checks and analysis-ready preparation.	3.0	Preprocessing progress and quality checks.	Peer review of preparation approach.
18 May	Helped plan NLP and author-reply network analysis sequence.	2.0	Analysis pipeline plan and Week 2 priorities.	Progress review and balanced workload confirmation.

Matteo Omizzolo (s4230379) - approximate weekly total: 9.5 hours.

Week 2: 19 May 2026 - 25 May 2026

Date range	Tasks completed	Approx. hours	Outputs produced	Collaboration / communication
19-20 May	Finalised preprocessing checks and contributed to network construction and metric review.	3.0	Validated interaction-network inputs and summaries.	Joint review of graph definitions.
21-22 May	Contributed to community, influence and diffusion interpretation; checked associated figures.	2.5	Network interpretation notes and figure checks.	Discussed findings with teammate.
23-24 May	Drafted/revised report material and checked consistency of results, tables and visuals.	3.0	Revised analysis narrative and QA notes.	Co-editing and results consistency review.
25 May	Supported presentation/final package review, README, filenames and repository organisation.	2.0	Submission package review completed.	Final joint checklist and approval.

Matteo Omizzolo (s4230379) - approximate weekly total: 10.5 hours.

Approximate total contribution for the project period: 20.0 hours.

3. Weekly Timesheets (Continued)

Team member: Emilija Milanovic (s4233176)

Week 1: 12 May 2026 - 18 May 2026

Date range	Tasks completed	Approx. hours	Outputs produced	Collaboration / communication
12-13 May	Confirmed research focus and contributed to shared scheduling and repository organisation.	2.0	Agreed project plan and workflow structure.	Joint planning and task allocation check-in.
14-15 May	Consolidated social-media text fields and reviewed data-cleaning requirements.	2.5	Preprocessing requirements and field checks.	Shared review of data quality decisions.
16-17 May	Worked on cleaned text preparation, language/duplicate checks and NLP-ready inputs.	3.0	Prepared text-processing progress and checks.	Peer review of cleaned-data readiness.
18 May	Helped define sentiment/topic and network-comparison analysis sequence.	2.0	Analysis pipeline plan and Week 2 priorities.	Progress review and balanced workload confirmation.

Emilija Milanovic (s4233176) - approximate weekly total: 9.5 hours.

Week 2: 19 May 2026 - 25 May 2026

Date range	Tasks completed	Approx. hours	Outputs produced	Collaboration / communication
19-20 May	Finalised preprocessing checks and contributed to sentiment/topic analysis and result review.	3.0	NLP summaries and validated text-analysis outputs.	Joint review of analytical settings.
21-22 May	Created/reviewed visualisations and contributed to cross-platform interpretation.	2.5	Figure/table checks and synthesis notes.	Discussed interpretation with teammate.
23-24 May	Drafted/revised report sections and reviewed alignment of notebooks, figures and narrative.	3.0	Revised reporting and consistency notes.	Co-editing and final results review.
25 May	Prepared/reviewed worksheet, access file, README, sample data and final repository package.	2.0	Submission documentation and packaging review.	Final joint checklist and approval.

Emilija Milanovic (s4233176) - approximate weekly total: 10.5 hours.

Approximate total contribution for the project period: 20.0 hours.

4. Individual Self-Reflection

Matteo Omizzolo (s4230379)

The strongest aspect of the project was the consistency of our collaboration across both analysis and writing. We maintained regular communication, checked progress as tasks developed and made interpretive decisions together rather than separating the work into isolated parts. My contribution included supporting data preparation, the interaction-network and diffusion analysis, interpretation of structural results, report revision and final quality review. Because the workload was divided evenly, I was also able to review NLP outputs and ensure that network findings were expressed consistently in figures and written conclusions. Minor issues, such as clarifying analysis definitions and aligning results across outputs, were handled through prompt discussion and peer checking. I learned that regular integration is particularly important in a project combining text and network methods. In a future project, I would establish a more formal shared log of decisions and completed checks from the beginning.

Emilija Milanovic (s4233176)

Our teamwork was effective because responsibilities were shared fairly and communication remained clear throughout the two-week period. We used regular check-ins to coordinate preprocessing, analysis, writing and final submission preparation, and significant choices about scope, results and presentation were made collaboratively. My contribution included preparing and reviewing text analysis inputs, contributing to sentiment/topic interpretation and visualisation, revising report content, and organising and reviewing the final repository documentation and submission files. I also checked alignment between the notebooks, figures and report narrative during final integration. Small coordination challenges were resolved constructively through discussion and mutual review rather than allowing inconsistencies to remain. This process reinforced the value of shared ownership of final outputs. For future work, I would schedule formal intermediate review milestones earlier so that final packaging requires fewer last-stage checks.

5. Rubric Alignment Check

The worksheet supports the Teamwork and Management criterion through the following documented features:

Criterion indicator	Evidence in this worksheet
Well-designed project plan	A dated plan sets out aim, workflow, responsibility sharing, quality control, risk management and final submission preparation.
Weekly plans	A two-week schedule identifies goals, tasks, expected outputs and communication procedures for 12-25 May 2026.
Realistic timesheets	Each member has specific, project-relevant activities and approximate hours, with equal 20.0-hour totals.
Clear management procedures	Communication, peer review, results checking, version coordination and package checks are documented.
Evidence of communication	Timesheets and plans record regular check-ins, joint decision-making and final approval procedures.
Even workload distribution	Both members contributed across analysis and final integration, with balanced approximate weekly totals.
Thoughtful individual reflection	Each reflection identifies contribution, collaborative practice, issue handling, learning and a practical improvement.